

RATON, NM, USA

WMO: 722678

Lat: 36.741N

Lon: 104.502W

Elev: 6349

StdP: 11.62

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 23052

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	4.1	8.7	-8.5	4.4	26.0	-4.3	5.5	33.4	35.1	35.6	31.0	38.5	6.8	50	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.9	92.9	58.7	90.2	58.4	87.6	58.2	63.9	80.9	62.7	79.7	61.8	78.6	12.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.3	95.6	67.1	58.0	91.2	66.3	56.7	87.2	66.0	32.4	80.9	31.5	79.0	30.7	78.1	67.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.1	25.7	21.9	DB	-7.5	96.4	5.7	2.4	-11.6	98.2	-14.9	99.6	-18.1	101.0	-22.2	102.7	(4)
(5)			WB	-8.2	66.0	5.5	1.2	-12.2	66.8	-15.4	67.5	-18.5	68.1	-22.6	68.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	32.2	34.2	42.4	49.1	57.8	67.8	71.7	69.3	63.0	51.3	39.9	31.3	
	DBStd	16.09	8.73	9.17	8.52	7.40	7.20	5.44	3.90	4.18	5.42	7.84	8.52	8.96	
	HDD50	2347	552	445	256	101	21	0	0	0	2	77	311	582	
	HDD65	5660	1017	864	700	477	237	33	3	11	95	424	753	1046	
	CDD50	2682	0	2	21	74	261	534	672	598	393	118	8	1	
	CDD65	521	0	0	0	0	13	117	210	144	36	1	0	0	
	CDH74	8208	0	0	19	77	593	2132	2588	1800	847	150	2	0	
	CDH80	3280	0	0	0	6	148	964	1192	708	247	15	0	0	
Wind	WSAvg	9.1	8.1	9.5	10.2	11.5	10.8	9.7	8.1	7.6	8.3	8.8	8.4	8.2	
Precipitation	PrecAvg	16.1	0.5	0.4	0.8	1.3	1.7	2.0	3.0	2.9	1.6	1.0	0.5	0.4	
	PrecMax	23.8	1.9	1.5	2.1	5.5	5.7	4.0	5.4	6.4	3.9	4.1	1.7	1.3	
	PrecMin	10.8	0.0	0.0	0.0	0.3	0.0	0.3	0.6	0.6	0.2	0.1	0.0	0.0	
	PrecStd	3.5	0.5	0.4	0.6	1.2	1.4	1.3	1.4	1.4	1.0	0.9	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	68.5	76.5	80.0	88.1	96.6	95.4	93.8	89.5	82.0	72.8	62.7	
		MCWB	41.6	44.0	46.7	48.4	53.1	57.1	60.8	59.6	57.5	51.8	46.4	41.7	
	2%	DB	58.8	62.9	72.1	76.1	84.1	92.9	92.8	90.2	85.8	78.3	68.3	57.9	
		MCWB	39.6	41.1	45.1	47.3	52.1	56.1	60.2	59.8	56.8	50.9	44.7	39.6	
	5%	DB	54.7	58.7	67.4	72.7	80.8	89.6	90.4	87.3	82.9	74.4	63.7	53.5	
		MCWB	37.8	39.2	43.4	46.4	51.4	56.1	59.9	59.9	56.2	49.9	42.8	37.7	
	10%	DB	50.2	53.5	62.6	69.0	76.9	86.0	87.5	84.0	79.5	69.9	58.3	48.6	
		MCWB	35.8	37.3	41.7	45.1	50.3	56.1	60.1	59.6	55.6	49.0	40.7	35.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	44.8	48.3	52.4	59.0	63.7	65.5	65.3	62.2	58.4	48.3	43.2
			MCDB	58.8	65.6	71.4	69.8	76.0	80.3	81.8	82.0	77.4	67.1	67.2	59.9
2%		WB	40.6	42.0	46.1	50.0	56.2	61.8	64.1	63.8	60.4	55.0	45.9	40.4	
		MCDB	56.8	61.2	68.8	68.0	72.7	79.4	81.4	80.4	75.2	67.8	65.1	55.6	
5%		WB	38.4	39.7	44.2	48.0	53.9	60.3	63.0	62.7	59.3	52.3	43.7	38.2	
		MCDB	53.4	56.9	65.3	67.2	71.5	78.2	80.7	78.8	73.6	68.1	61.8	52.3	
10%		WB	36.2	37.4	42.2	46.2	52.3	59.0	62.0	61.6	58.1	50.2	41.4	35.7	
		MCDB	49.4	52.2	60.7	65.9	70.8	76.9	79.7	77.2	71.6	66.4	57.6	48.1	
Mean Daily Temperature Range	5% DB	MDBR	30.7	30.2	32.4	31.8	31.6	32.9	30.9	29.5	30.3	31.6	31.7	28.8	
		MCDBR	37.6	39.3	40.9	39.5	37.9	38.6	35.8	34.2	36.3	39.5	39.0	35.5	
		MCWBR	21.6	20.9	19.4	17.7	15.0	13.6	10.7	11.0	13.0	17.2	19.8	20.3	
	5% WB	MCDBR	35.4	36.6	38.3	34.8	32.4	31.7	29.8	28.7	28.8	31.5	36.6	33.4	
		MCWBR	20.7	19.7	18.7	16.1	13.6	12.3	9.7	10.0	11.1	14.6	18.9	19.6	
Clear-Sky Solar Irradiance	taub	0.233	0.239	0.267	0.288	0.301	0.323	0.356	0.340	0.299	0.273	0.246	0.237		
	taud	2.630	2.632	2.547	2.493	2.518	2.455	2.443	2.474	2.556	2.601	2.607	2.606		
	Ebn at Noon	316	325	320	316	311	302	291	295	304	306	307	306		
	Edh at Noon	22	25	30	34	33	36	36	34	30	25	22	21		
All-Sky Solar Radiation	RadAvg	975	1272	1669	2030	2265	2368	2190	2014	1788	1410	1059	850		
	RadStd	62	88	114	120	152	107	85	92	97	85	64	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144	+83

Nomenclature: See separate page